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Using device: mps
Using single-amplitude feyn data from /var/folders/1s/69py7bdx0g7g0t4_8rjh6g9h0000gn/T/single_amplit
ude_eval_7o6pubns/first_amplitude.csv
Imported 1 token rows
Wrote raw expression data to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML
/data_testing/outputs/ym_4pt_colour_500k_beam20.csv
Wrote normalized token data to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-
ML/data_testing/outputs/ym_4pt_colour_500k_beam20_tok.csv
Input token limit: unlimited; longest input: 1247 content / 1249 with BOS/EOS
Numerical evaluator: ym (energy_scale=2.0; pol_modes=coulomb,covariant; tol_abs=1e-10; tol_rel=1e-08
)
Loading model from /Users/kymani/Documents/Big_YM_Data_Sets/Kymani_Colour_Models/4POINT_YM_Model_500
k_Colour/best_model.pt

Running decode mode: beam
  method      : beam
  max_length  : 2000
  beam_size   : 20
  wrote human CSV to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_te
sting/outputs/ym_4pt_colour_500k_beam20_beam_human.csv

[beam] n=1
  top-1 correct   :    1 / 1 (100.0%)
  any-beam correct:    1 / 1 (100.0%)
  avg valid beams : 14.0
  avg tokens (correct): pred 87.0 vs scrambled 1247.0

Example predictions (1 rows):
  row 0 [beam]
    target simple : (((e_1 · e_2)*(e_3 · e_4)*(p_1 · p_3))/(p_1 · p_2)) + (((e_1 · e_2)*(e_3 · p_1)*
(e_4 · p_1)*(p_3 · p_4))/((p_1 · p_2)*(p_1 · p_4))) - (((e_1 · e_2)*(e_3 · p_1)*(e_4 · p_3))/(p_1 ·
input scram   : (((e_1 · e_2)*(e_3 · e_4)*(p_1 · p_3))/(p_1 · p_2)) + (((e_1 · e_2)*(e_3 · p_1)*
(e_4 · p_1)*(p_3 · p_4))/((p_1 · p_2)*(p_1 · p_4))) - (((e_1 · e_2)*(e_3 · p_1)*(e_4 · p_3))/(p_1 ·
top1 pred     : (p_1·F_2·F_3·p_1·p_2·F_1·F_4·p_2)/(((p_1·p_2))^2·p_1·p_3·p_1·p_4) + (p_1·F_2·F_4
·p_1·p_2·F_1·F_3·p_2)/(((p_1·p_2))^2·((p_1·p_4))^2) + (p_1·F_3·F_4·p_1·p_3·F_1·F_2·p_3)/(p_1·p_2·p_1
selection      : numerically_equivalent_rerank
top1 metrics   : token=0 string=0 num_simple=1 num_scr=1
any-beam       : token=0 string=0 num_simple=1 num_scr=1
  wrote detail results to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/da
ta_testing/outputs/ym_4pt_colour_500k_beam20_beam_results.csv

Wrote summary to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_testin
g/outputs/ym_4pt_colour_500k_beam20_summary.csv
Total wall time: 4.07s

Terminal output PDF: /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_te
sting/outputs/Yang-mills_4_Evaluation_Information_2026-09-21_00-22-29-846271.pdf

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